Dataset Link:

Chest X-Ray Images (Pneumonia)

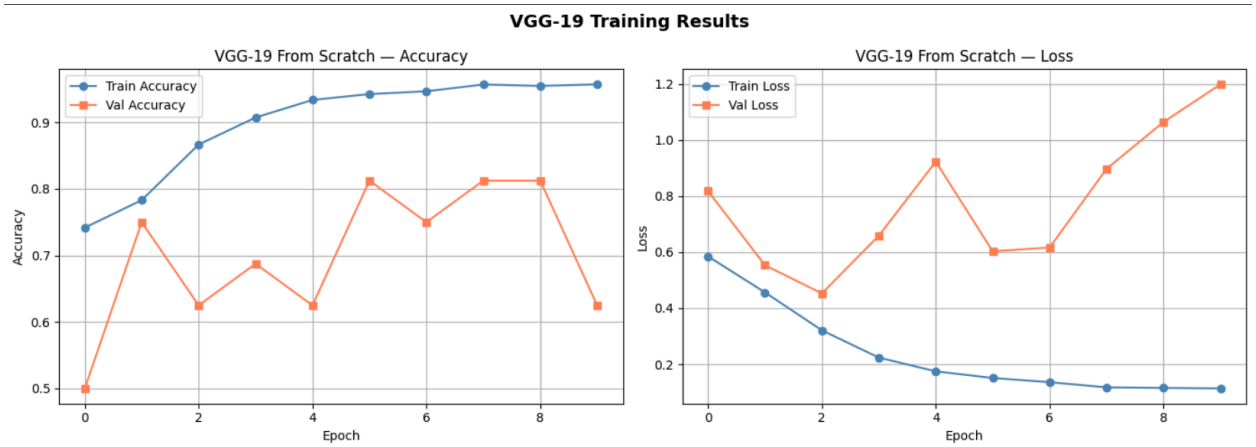
Results And Figures:

VGG-19 Model Summary:
Total params: 139,578,434 (532.45 MB)
Trainable params: 139,578,434 (532.45 MB)
Non-trainable params: 0 (0.00 B)

Layer (type)	Output Shape	Param #
conv2d_16 (Conv2D)	(None, 224, 224, 64)	1,792
conv2d_17 (Conv2D)	(None, 224, 224, 64)	36,928
max_pooling2d_5 (MaxPooling2D)	(None, 112, 112, 64)	0
conv2d_18 (Conv2D)	(None, 112, 112, 128)	73,856
conv2d_19 (Conv2D)	(None, 112, 112, 128)	147,584
max_pooling2d_6 (MaxPooling2D)	(None, 56, 56, 128)	0
conv2d_20 (Conv2D)	(None, 56, 56, 256)	295,168
conv2d_21 (Conv2D)	(None, 56, 56, 256)	590,080
conv2d_22 (Conv2D)	(None, 56, 56, 256)	590,080
conv2d_23 (Conv2D)	(None, 56, 56, 256)	590,080
max_pooling2d_7 (MaxPooling2D)	(None, 28, 28, 256)	0
conv2d_24 (Conv2D)	(None, 28, 28, 512)	1,180,160
conv2d_25 (Conv2D)	(None, 28, 28, 512)	2,359,808
conv2d_26 (Conv2D)	(None, 28, 28, 512)	2,359,808
conv2d_27 (Conv2D)	(None, 28, 28, 512)	2,359,808
max_pooling2d_8 (MaxPooling2D)	(None, 14, 14, 512)	0
conv2d_28 (Conv2D)	(None, 14, 14, 512)	2,359,808
conv2d_29 (Conv2D)	(None, 14, 14, 512)	2,359,808
conv2d_30 (Conv2D)	(None, 14, 14, 512)	2,359,808
conv2d_31 (Conv2D)	(None, 14, 14, 512)	2,359,808
max_pooling2d_9 (MaxPooling2D)	(None, 7, 7, 512)	0
flatten_1 (Flatten)	(None, 25088)	0
dense_3 (Dense)	(None, 4096)	102,764,544

Layer (type)	Output Shape	Param #
dropout_2 (Dropout)	(None, 4096)	0
dense_4 (Dense)	(None, 4096)	16,781,312
dropout_3 (Dropout)	(None, 4096)	0
dense_5 (Dense)	(None, 2)	8,194

VGG-19 Training Results:



Evaluating On Test Set:

VGG-19 Test Accuracy: 0.7676
VGG-19 Test Loss: 0.7357

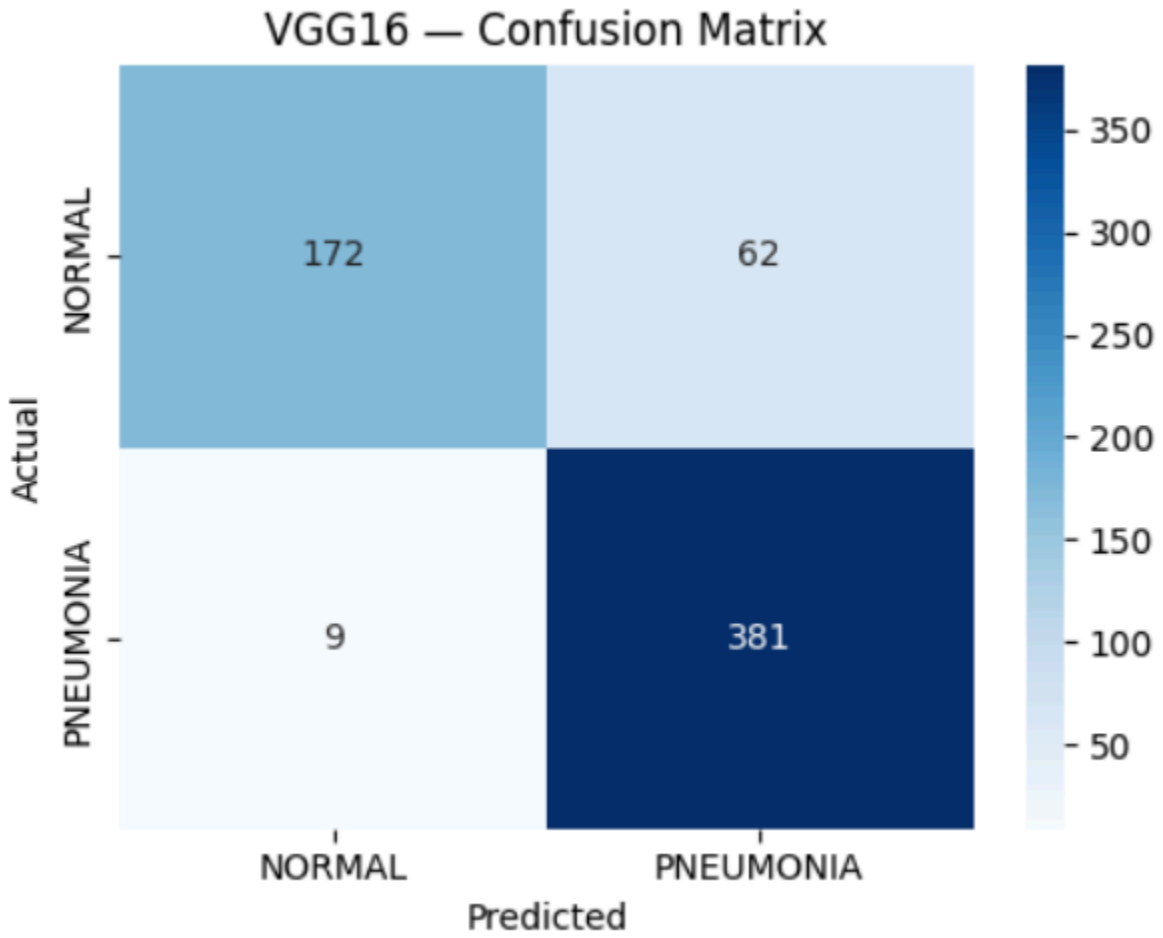
Results of 3 Classification Models:

1. VGG-16

Class	Precision	Recall	F1-Score	Support
NORMAL	0.95	0.74	0.83	234
PNEUMONIA	0.86	0.98	0.91	390
Accuracy			0.89	624
Macro Avg	0.91	0.86	0.87	624

Class	Precision	Recall	F1-Score	Support
Weighted Avg	0.89	0.89	0.88	624

Confusion Matrix:

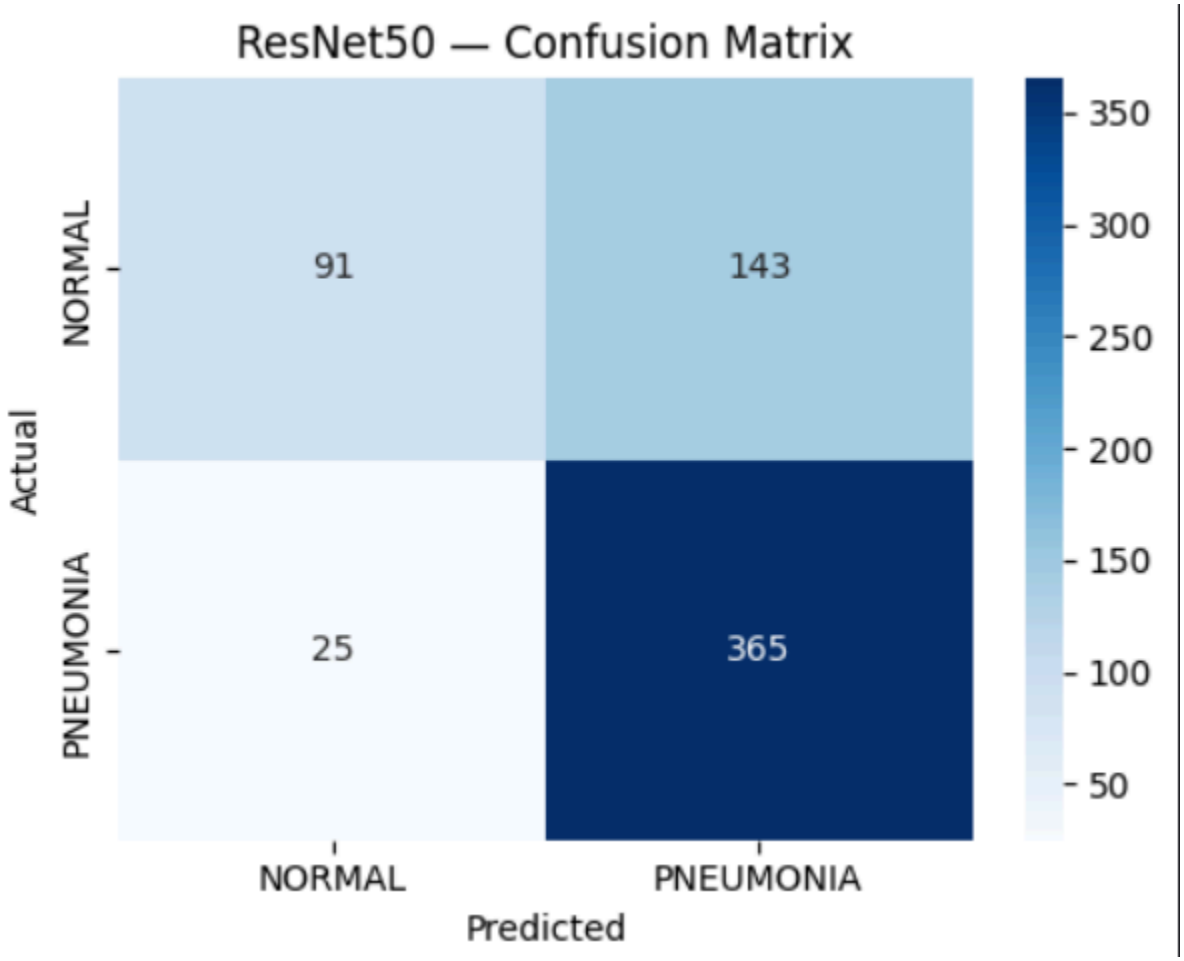


2. ResNet-50

Class	Precision	Recall	F1-Score	Support
NORMAL	0.78	0.39	0.52	234
PNEUMONIA	0.72	0.94	0.81	390
Accuracy			0.73	624

Class	Precision	Recall	F1-Score	Support
Macro Avg	0.75	0.66	0.67	624
Weighted Avg	0.74	0.73	0.70	624

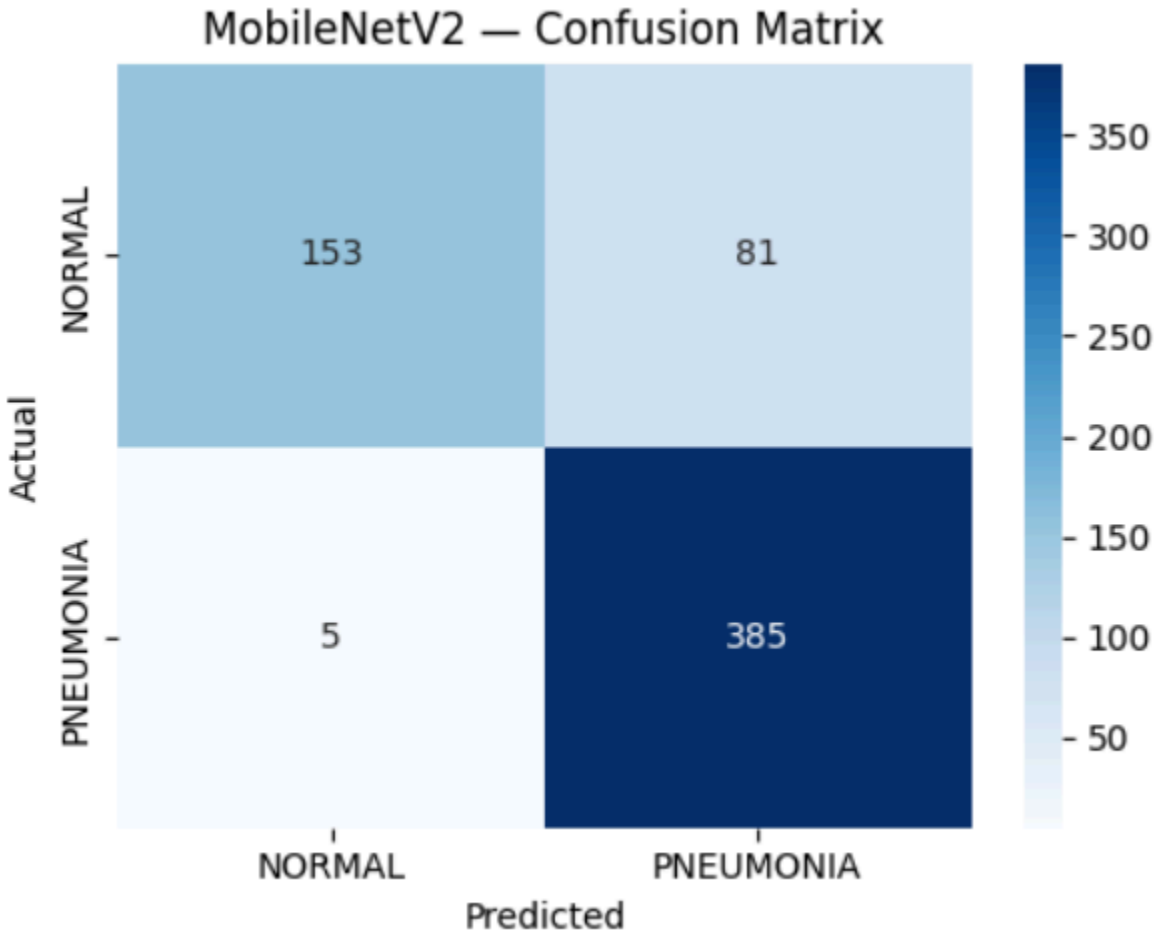
Confusion Matrix:

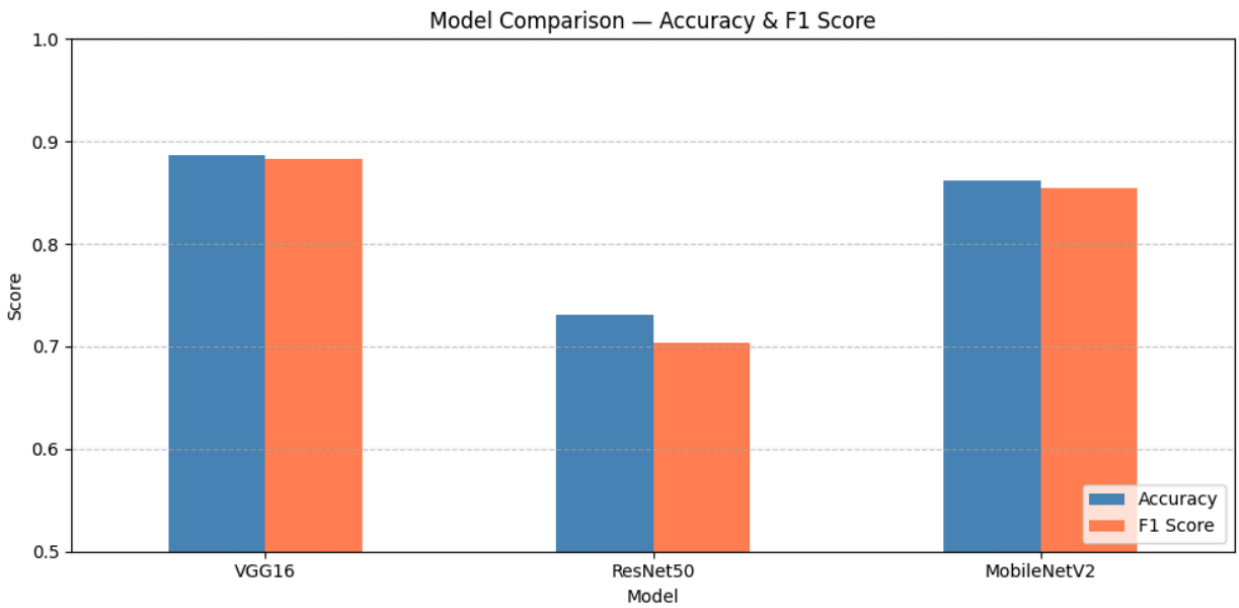


3. **MobileNetV2**

Class	Precision	Recall	F1-Score	Support
NORMAL	0.97	0.65	0.78	234
PNEUMONIA	0.83	0.99	0.90	390
Accuracy			0.86	624
Macro Avg	0.90	0.82	0.84	624
Weighted Avg	0.88	0.86	0.85	624

Confusion Matrix:



Final Comparison Among Models:

Model	Accuracy	F1 Score
VGG16	88.62%	88.26%
MobileNetV2	86.22%	85.49%
ResNet50	73.08%	70.31%